<table>
<tr><th>Question</th><th>Marking Guidance</th><th>Mark</th><th>Comments</th></tr>
<tr>
<td>07.1</td>
<td>1. Variety and D and E in left column;
2. Time and ‘/ min’ and Mean (time) ‘/ min’ in column heading only;
3. Correctly calculated times and means calculated to one decimal place;
Examples achieving 3 marks
<table>
<tr><th>Variety</th><th>Time / minutes</th><th>Mean / minutes</th></tr>
<tr><td rowspan="3">D</td><td>15.8/15.83/16</td><td rowspan="3">16.1 OR 16.2 OR 16.3</td></tr>
<tr><td>18</td></tr>
<tr><td>14.5/15</td></tr>
<tr><td rowspan="3">E</td><td>6.5/7</td><td rowspan="3">7.2 OR 7.3</td></tr>
<tr><td>8</td></tr>
<tr><td>7</td></tr>
</table>
<table>
<tr><th>Apple</th><th>Time / s</th><th>Mean / s</th></tr>
<tr><td>D</td><td>950</td><td></td></tr>
<tr><td></td><td>1080</td><td>966.7</td></tr>
<tr><td></td><td>870</td><td></td></tr>
<tr><td>E</td><td>390</td><td></td></tr>
<tr><td></td><td>480</td><td>430.0</td></tr>
<tr><td></td><td>420</td><td></td></tr>
</table>
</td>
<td>3
(3 x AO2)</td>
<td>Accept if no table lines are shown, figures clearly written in rows and columns
Accept tables where mean is given in middle column and time is given in right column
1. Accept type OR apple for variety
1. Accept D and E in either order
2. Accept mean values shown in rows rather than in columns
2. Accept brackets OR “in” for solidus
2. Ignore average for mean
2. Accept min OR mins for minutes
2. Accept seconds OR secs OR sec for seconds
2. Reject mixed units eg “/ minutes and seconds”
3. Accept correct times and correct means however presented</td>
</tr>
</table>

Question	Marking Guidance	Mark	Comments
07.2	1. More phenol/substrate in E **OR** More PPO/enzyme in E **OR** Higher PPO/enzyme activity in E; 2. (So) more enzyme-substrate/E-S complexes (form);	2 (2 x AO2)	1. and 2. Accept the converse for D 1. Ignore oxygen 2. Ignore ESC

Question	Marking Guidance	Mark	Comments
07.3	1. Use (enzyme's) higher/optimum temperature; 2. More kinetic energy **OR** (So) more active enzyme; 3. (Causing) increased rate of reaction **OR** (Causing) more enzyme-substrate complexes/collisions; **OR** 4. Use (enzyme's) lower/optimum temperature **OR** Use (enzyme's) optimum pH; 5. Less denaturation; **OR** (So) more active enzyme; 6. (Causing) increased rate of reaction **OR** (Causing) more enzyme-substrate complexes/collisions; **OR** 7. Crush/grind/blend/chop the apple (tissue);	3 max (3 x AO3)	Mark in groups, 1, 2 and 3. **OR** 4, 5 and 6 **OR** 7, 8 and 9 3. 6. and 9. Accept E-S complexes for enzyme-substrate complexes 3. 6. and 9. Accept a description of increased rate, eg 'speeds up reaction' 1 and 4 Accept change in temperature 7. Accept cube for apple

	8 (So) more PPO/phenol/substrate in contact with oxygen/air; 9. (Causing) increased rate of reaction **OR** (Causing) more enzyme-substrate complexes/collisions;		7. Accept dice **OR** homogenise **OR** cut up **OR** slice for crush If oxygen is suggested, no mark for oxygen but MP8 and MP9 are available

Question	Marking Guidance	Mark	Comments
07.4	Measure the intensity of brown colour by comparing apple tissue with a colour chart showing a range of apple tissues of known pigment concentration.	1 (1 x AO3)	